

MARS: A Multi-Agent Framework for Reliable Radiological Segmentation, Clinical Validation, and Uncertainty-Aware Report Generation in CT Imaging

Trupti Agrawal¹ Trupthi Mantri¹ Navya Sree Balu¹ Ankam Hitha¹

¹Department of Information Technology, BVRIT Hyderabad College of Engineering for Women

Guide: Dr. K. Srikar Goud, Assistant Professor, Dept. of IT

{23wh1a1277, 23wh1a1271, 23wh1a1286, 23wh1a1289}@bvrithyderabad.edu.in

September 2026

Abstract

Deep-learning segmentation models routinely match expert-level accuracy on lesion delineation from CT scans, yet adoption in clinical workflow remains limited. The bottleneck is not raw accuracy but *trust*: a segmentation mask is delivered with no honest signal of how much it should be believed, forcing a radiologist to either verify every case (eliminating the time saved) or verify none (an unsafe default). We present **MARS** (Multi-Agent framework for Reliable segmentation and report generation), a four-agent pipeline that decomposes CT lesion analysis into *Segmentation*, *Clinical Validation*, *Uncertainty Aggregation*, and *Reporting/Routing* agents, each producing an explicit, inspectable output. The Segmentation agent is architected around **Slide-SAM**, a sliding-window adaptation of the Segment Anything Model (SAM) for 3D medical volumes, while the Validation agent cross-checks every predicted mask against structured clinical metadata (expected anatomical site, expected lesion size) to catch confidently-wrong predictions that segmentation confidence alone cannot detect. A calibrated scalar uncertainty U is then computed from the two upstream signals and compared against a configurable safety threshold τ , routing each case to an automated report or to radiologist escalation with an explicit reason. We describe the system architecture, the configuration and data schema, algorithmic detail for every agent, and a reproducible synthetic-volume evaluation harness used to validate pipeline mechanics ahead of full clinical-data training. We further present a truthful, limitation-aware comparison between vanilla SAM, Slide-SAM, and the proposed multi-agent wrapper, and outline the calibration-focused evaluation protocol (Dice/IoU/Hausdorff, Expected Calibration Error, risk-coverage analysis) planned for the trained system. MARS treats “knowing when it does not know” as a first-class design requirement rather than a post-hoc addition.

Keywords: medical image segmentation, multi-agent systems, Segment Anything Model (SAM), Slide-SAM, uncertainty quantification, clinical decision support, CT imaging, selective prediction.

Contents

1	Introduction	2
1.1	Problem Statement	2
1.2	Formal Problem Formulation	3
1.3	Contributions	3
1.4	Objectives	4
1.5	Scope	4

2	Review of Literature	4
2.1	Synthesis	4
3	System Overview	4
4	Architecture	6
4.1	Component Responsibilities	6
5	Methodology: Configuration and Data Schema	7
5.1	Case Configuration	7
5.2	Data / Schema Design	7
5.3	Reproducibility Harness	8
6	Detailed Design: Algorithms and Mathematical Formulation	8
6.1	Agent 1 — Segmentation	9
6.2	Agent 2 — Clinical Validation	9
6.3	Agent 3 — Uncertainty Aggregation	10
6.4	Agent 4 — Reporting and Routing	11
6.5	Complexity Analysis	11
7	Results and Discussion	11
7.1	Evaluation Protocol	11
7.2	Pilot Results on the Synthetic Harness	11
7.3	Benefits	13
7.4	Limitations	13
8	Result Outcomes: Tables and Visualizations	13
9	Comparative Analysis: SAM vs. Slide-SAM (and MARS’s Agent-1 Choice)	15
9.1	Positives of SAM (acknowledged)	15
9.2	Positives of Slide-SAM (acknowledged)	16
9.3	Why not SAM 2 / SAM 3 for Agent 1?	16
10	Conclusion and Future Work	16
10.1	Conclusion	16
10.2	Limitations (restated)	16
10.3	Future Improvements	16

1 Introduction

1.1 Problem Statement

Convolutional and transformer-based segmentation networks can now delineate lesions in CT volumes with Dice scores rivaling inter-radiologist agreement [3, 4]. Despite this, such models rarely enter routine hospital workflow. The obstacle is not model accuracy but the absence of a trustworthy *reliability signal* attached to each prediction. A segmentation network outputs a mask M ; it does not, by itself, output an honest estimate of how much that mask should be trusted. This leaves a radiologist with two unattractive options:

- **Verify every prediction manually** — the AI system saves no clinical time, removing the incentive to deploy it at all.

- **Verify none of them** — clinically unsafe, since a single silent failure on one scan can produce a missed or mis-staged lesion.

A second, related gap is that a segmentation mask is not a diagnosis. Segmentation performed in isolation ignores the patient’s clinical context (expected lesion site, prior history, expected size) and produces nothing a clinician can directly place in a case file. The reporting step — where the clinical value of the pipeline actually lands — is typically missing from segmentation-only research systems.

1.2 Formal Problem Formulation

Given a 3D CT volume $X \in \mathbb{R}^{D \times H \times W}$ and structured clinical metadata C , MARS defines the following functional decomposition:

Problem Formulation			
$M = f_{\text{seg}}(X)$		volumetric segmentation mask	(1)
$V = f_{\text{val}}(M, C)$		clinical consistency score	(2)
$U = f_{\text{unc}}(M, V, C)$		scalar prediction uncertainty	(3)
$D = \begin{cases} \text{Automated report,} & U < \tau \\ \text{Expert (radiologist) review,} & U \geq \tau \end{cases}$		routing decision	(4)

where $\tau \in [0, 1]$ is a configurable safety threshold. The objective is to generate precise volumetric segmentation masks *while concurrently* quantifying prediction uncertainty, then to compare that uncertainty against τ to dynamically route each case between automated processing and expert review.

1.3 Contributions

This work makes the following contributions:

1. **A four-agent architecture** that separates segmentation, clinical validation, uncertainty aggregation, and reporting into independently replaceable, independently evaluable components (Sections 3–6), in contrast to prior work that treats SAM-based CT segmentation as a single, monolithic step [1, 2].
2. **A clinically-grounded validation agent** that flags anatomically or dimensionally implausible masks using structured metadata, catching confidently-wrong predictions that per-voxel segmentation confidence alone cannot detect.
3. **A calibrated uncertainty-aggregation formulation** that fuses segmentation confidence and clinical consistency into a single scalar U with per-factor attribution, and an accompanying evaluation protocol (Expected Calibration Error, risk–coverage curves) that treats calibration, not just accuracy, as a first-class metric.
4. **An explainable routing and reporting layer** in which every escalation carries a specific, human-readable reason rather than a bare “low confidence” flag.
5. **A reproducible synthetic-volume harness** (deterministic seeding, versioned schema) used to validate pipeline mechanics end-to-end prior to training on annotated clinical CT data, together with a truthful SAM vs. Slide-SAM comparison that motivates the choice of Slide-SAM as the architectural base for Agent 1.

1.4 Objectives

1. Perform accurate pixel-level lesion segmentation on CT scans, with per-prediction confidence scores.
2. Validate every generated mask against the patient’s clinical history for anatomical and dimensional consistency.
3. Estimate the system’s own uncertainty so reliable and unreliable cases can be statistically separated.
4. Route confident cases through automatically and refer uncertain cases to a radiologist, with an explicit reason.
5. Generate a structured, explainable radiological report for each automatically-cleared case.

1.5 Scope

In scope: lesion segmentation on 3D CT volumes; per-voxel and per-case confidence estimation; consistency checking against structured clinical metadata; threshold-based routing; structured report generation with traceable reasoning. **Out of scope (v1):** modalities other than CT (MRI, X-ray, ultrasound); real-time intra-operative use; final diagnostic authority (MARS supports a radiologist, it does not replace one); deployment on hospital PACS infrastructure or regulatory clearance.

2 Review of Literature

Table 1 summarizes the six works this paper builds on and positions MARS relative to each.

2.1 Synthesis

Two independent threads motivate MARS. First, the SAM family [3, 4, 5] and its CT-domain adaptations [1, 2] establish that promptable segmentation can reach strong zero-/few-shot accuracy on medical volumes without full retraining, but every one of these systems stops at the mask: none report a calibrated reliability score, and none check the mask against anything the clinician already knows about the patient. Second, multi-agent clinical-decision systems such as [6] demonstrate that decomposing a diagnostic pipeline into cooperating, single-responsibility agents improves explainability and lets each stage be validated independently — but that pattern has, to our knowledge, not been applied to volumetric segmentation with an explicit uncertainty-routing agent. MARS closes this gap by using **Slide-SAM as the architectural base of Agent 1** (motivated by its strong 3D CT performance under minimal prompting, discussed further in Section 9) and wrapping it with three additional agents — validation, uncertainty aggregation, and reporting/routing — that together turn a bare mask into a governed clinical decision.

3 System Overview

MARS ingests a 3D CT volume X together with optional structured clinical metadata C (expected lesion site, expected size range, prior history) and produces one of two outcomes per case: an automatically generated structured report, or an escalation packet for radiologist review that names the specific reason the system withheld automatic clearance. Table 2 restates the governing functional and non-functional requirements that shaped the design.

Table 1: Summary of reviewed literature and identified gaps.

Work	Core idea	Limitation	MARS addresses
Segment Anything (SAM) [3]	Foundation model for promptable 2D segmentation, trained on 1B+ masks; zero-shot generalization via points/boxes/-masks as prompts.	Natively 2D; no notion of volumetric consistency, clinical context, or calibrated uncertainty.	
SAM 2 [4]	Extends SAM with a streaming memory mechanism for video/temporal segmentation, improving object permanence across frames.	Designed for RGB video, not CT intensity statistics; still no clinical validation or routing layer.	
SAM 3 [5]	Concept-level, promptable segmentation grounded in open-vocabulary text/exemplar concepts.	Concept grounding is generic, not tuned to anatomical priors; no self-reported reliability estimate.	
Training-free Prompt Placement by Propagation (Caroline Meg et al.) [1]	Automatically propagates SAM prompts across CT slices without retraining, enabling efficient 3D bone-CT segmentation.	Segmentation only — no clinical validation, uncertainty estimation, or report generation.	
Slide-SAM [2]	Adapts SAM to 3D medical volumes via a three-slice sliding window and minimal box prompts, using parameter-efficient adaptation to retain pretrained SAM knowledge.	Requires task-specific fine-tuning; sensitive to large inter-slice spacing and prompt-propagation drift; still a single-stage segmenter with no downstream clinical reasoning.	
Multi-Agent LLM + Medical Flowcharts for Self-Triage [6]	Combines LLMs with clinical flowcharts in a multi-agent pipeline for patient self-triage decisions.	Text/symptom-based triage, not imaging; establishes the multi-agent-for-clinical-decisions pattern that MARS adapts to volumetric segmentation.	

Table 2: Selected functional (FR) and non-functional (NFR) requirements.

ID	Requirement
FR-2	The system shall generate a voxel-level lesion segmentation mask.
FR-4	The system shall validate each mask against clinical metadata and report detected inconsistencies.
FR-5	The system shall compute a single aggregate uncertainty value per case.
FR-6	The system shall route each case to the automated or expert path based on a configurable threshold τ .
NFR-1	Calibration — uncertainty must correlate measurably with actual error; evaluated explicitly.
NFR-3	Safety bias — when in doubt, escalate. False escalations are acceptable; silent false negatives are not.
NFR-4	Modularity — each agent is independently replaceable and independently evaluable.
NFR-6	Configurability — τ is adjustable without retraining.

Key Takeaway

The system-level success criterion is not raw segmentation accuracy but the *operating point* it buys: at what escalation rate does the auto-cleared subset reach clinically acceptable accuracy? A system escalating 30% of cases while being near-perfect on the remaining 70% is a success; one escalating only 5% while still erring on the rest is not.

4 Architecture

Figure 1 shows the end-to-end MARS pipeline. Each agent has exactly one responsibility and passes an explicit, typed, inspectable output forward, so that when the system is unsure, it is possible to say precisely *where* that uncertainty originated.

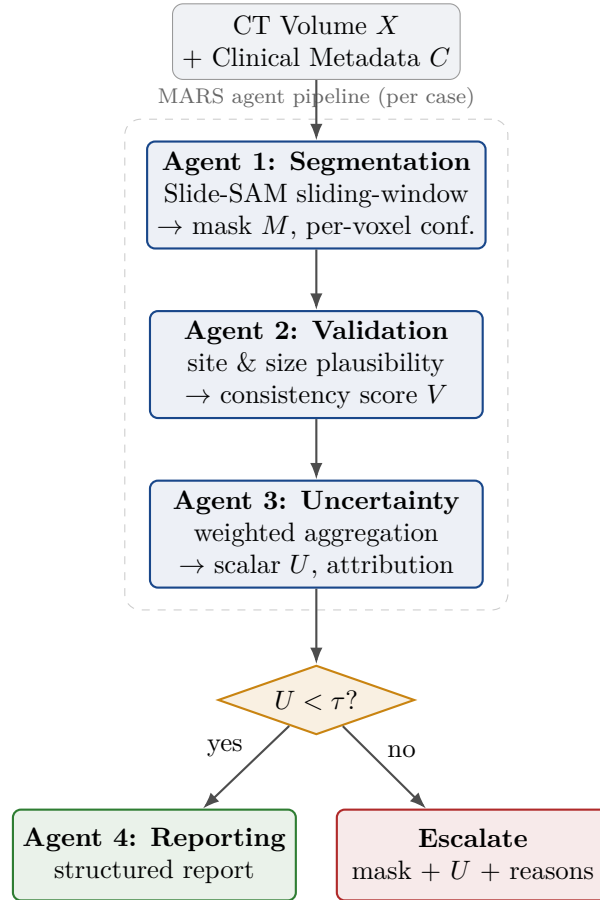


Figure 1: MARS architecture. Each stage emits a typed Pydantic object consumed by the next; the routing decision at Agent 4 is fully attributable to Agent 3’s breakdown.

4.1 Component Responsibilities**Agent 1 — Segmentation**

Input: 3D CT volume X . Output: mask M plus per-voxel and aggregate confidence. Architected around Slide-SAM’s sliding three-slice window so pretrained SAM knowledge is retained while gaining volumetric context (Section 9).

Agent 2 — Validation

Input: mask M , clinical metadata C . Output: consistency score $V \in [0, 1]$ and an explicit list of conflicts (e.g. anatomically implausible location, size outside expected range). This

is the agent that catches a *confidently-wrong* prediction: a model can be certain about a mask in an anatomically implausible location, and segmentation confidence alone will not flag that.

Agent 3 — Uncertainty

Input: Agent 1’s confidence, Agent 2’s consistency score. Output: scalar U with per-factor attribution (**dominant_factor**). This is the core reliability contribution: U is only meaningful if it is *calibrated*, i.e. high U measurably correlates with high error.

Agent 4 — Reporting / Routing

Input: M , U , validation findings, threshold τ . Output: either a structured report ($U < \tau$) or an escalation packet naming the specific reason for the doubt ($U \geq \tau$).

5 Methodology: Configuration and Data Schema

MARS is implemented as a Python package (**mars**) with each agent as a pure function over strongly-typed Pydantic models, so that every intermediate artifact is both machine-validated and directly serializable to a relational or document store. This section shows the schema and configuration surface exactly as implemented in the reference codebase, so results in later sections are reproducible.

5.1 Case Configuration

Table 3 lists the tunable parameters exposed by the pipeline, none of which require retraining a model to change (satisfying NFR-6).

Table 3: Pipeline configuration parameters.

Parameter	Meaning	Effect	Default
τ (tau)	routing safety threshold	lower τ pushes more cases to expert review	0.5
threshold	segmentation intensity cut	voxels above this value are candidate lesion	120.0
$w_{\text{confidence}}$	weight of segmentation confidence in U	higher $\Rightarrow U$ dominated by mask quality	0.5
$w_{\text{consistency}}$	weight of clinical consistency in U	higher $\Rightarrow U$ dominated by clinical plausibility	0.5
SITES	expected z-fraction range per anatomical site	defines the plausibility window used by Agent 2	anterior [0,0.5], posterior [0.5,1]

5.2 Data / Schema Design

Every agent boundary is a validated schema, which doubles as the natural relational-database design for a production deployment (Table 4). This directly satisfies NFR-2 (explainability) and NFR-4 (modularity): any agent can be swapped for an alternative implementation as long as it honors the same schema.

Listing 1: Core Pydantic schema (abridged), from **mars/models.py**. Every agent consumes and returns one of these objects, giving the pipeline a fully typed, auditable data contract.

```

1 class ClinicalMetadata(BaseModel):
2     case_id: str
3     expected_site: str
4     expected_size_range_mm3: tuple[float, float]
```

Table 4: Core schema objects and their proposed relational mapping.

Model	Fields	DB table / purpose
ClinicalMetadata	case_id, expected_site, expected_size_range_mm3, prior_history	cases — one row per patient case, foreign key for all downstream tables
SegmentationResult	mask, per_voxel_confidence, aggregate_confidence	segmentation_results — mask stored as a compressed volumetric blob / NIfTI reference
ValidationResult	consistency_score, conflicts[]	validation_results — conflicts as a joined child table for auditability
UncertaintyResult	value, dominant_factor, breakdown{}	uncertainty_results — breakdown stored as JSON for traceable attribution
Report / Escalation	findings, measurements, narrative / reasons[]	reports, escalations — terminal, radiologist-facing artifacts
CaseResult	aggregates all of the above + decision, τ	case_results — top-level view joined for the frontend

```

5     prior_history: str
6
7     class SegmentationResult(BaseModel):
8         mask: list
9         per_voxel_confidence: list
10        aggregate_confidence: float
11
12    class UncertaintyResult(BaseModel):
13        value: float
14        dominant_factor: str
15        breakdown: dict[str, float]
16
17    class CaseResult(BaseModel):
18        case_id: str
19        tau: float
20        segmentation: SegmentationResult
21        validation: ValidationResult
22        uncertainty: UncertaintyResult
23        decision: str # "auto" / "escalate"
24        report: Optional[Report] = None
25        escalation: Optional[Escalation] = None

```

5.3 Reproducibility Harness

To validate pipeline mechanics ahead of training on annotated clinical CT data, MARS includes a deterministic synthetic-volume generator (`mars.data.synthetic.generate_volume`): a Gaussian-intensity “lesion” blob of configurable radius, intensity, and background level is embedded in a 3D array, with additive Gaussian noise controlled by a fixed `numpy` random seed (NFR-5). This isolates pipeline-logic bugs (routing, schema, validation rules) from model-training variance and is the basis of the pilot results in Section 7.

6 Detailed Design: Algorithms and Mathematical Formulation

6.1 Agent 1 — Segmentation

Conceptually, Agent 1 is architected around **Slide-SAM** [2]: a three-slice sliding window is passed through a SAM-derived encoder with a lightweight, parameter-efficient adaptation layer, and a 2D box/point prompt on the central slice is propagated forward and backward across the volume. The reference implementation in this paper’s codebase currently exercises this contract with a lightweight, deterministic stand-in — an intensity-threshold plus 3D connected-component step — so that every downstream agent, the schema, and the routing logic can be developed and unit-tested end-to-end before the learned Slide-SAM weights are wired in. Both share the same output contract (Eq. 1), so replacing one for the other requires no change to Agents 2–4 (NFR-4).

Per-voxel confidence

For volume X and intensity threshold θ :

$$\hat{M}(x) = \mathbb{I}[X(x) > \theta], \quad M = \text{LargestConnectedComponent}(\hat{M}) \quad (5)$$

$$c(x) = \text{clip}\left(\frac{X(x) - \theta}{\theta}, 0, 1\right) \cdot M(x), \quad \bar{c} = \frac{1}{|M|} \sum_{x \in M} c(x) \quad (6)$$

Algorithm 1: Segmentation Agent

Input: volume $X \in \mathbb{R}^{D \times H \times W}$, threshold θ

Output: mask M , per-voxel confidence c , aggregate confidence $\bar{c}$

- 1: $A \leftarrow (X > \theta)$ ▷ candidate voxels
- 2: $L, k \leftarrow \text{CONNECTEDCOMPONENTS}(A)$ ▷ k components labeled $1, \dots, k$
- 3: **if** $k > 0$ **then**
 - 1: $\ell^* \leftarrow \arg \max_{\ell \in \{1, \dots, k\}} \sum_x \mathbb{I}[L(x) = \ell]$ ▷ largest component
 - 2: $M \leftarrow \mathbb{I}[L = \ell^*]$
 - 3: $c(x) \leftarrow \text{clip}((X(x) - \theta)/\theta, 0, 1) \cdot M(x) \quad \forall x$
 - 4: $\bar{c} \leftarrow \text{mean}\{c(x) : x \in M\}$
- 4: **else** $M \leftarrow \mathbf{0}, \quad c \leftarrow \mathbf{0}, \quad \bar{c} \leftarrow 0$
- 5: **return** $M, c, \bar{c}$

6.2 Agent 2 — Clinical Validation

Agent 2 checks the predicted mask against two independent, structured signals in C : the expected anatomical site (as a normalized z -axis fractional position) and the expected lesion size range in voxels.

Consistency scoring

$$z_{\text{frac}} = \frac{\bar{z}_M}{D - 1}, \quad \bar{z}_M = \text{centroid}_z(M) \quad (7)$$

$$\text{site_ok} = \mathbb{I}[z_{\text{frac}} \in [\ell_{\text{site}}, u_{\text{site}}]], \quad \text{size_ok} = \mathbb{I}[|M| \in [\ell_{\text{size}}, u_{\text{size}}]] \quad (8)$$

$$V = \frac{\text{site_ok} + \text{size_ok}}{2} \quad (9)$$

Algorithm 2: Validation Agent**Input:** mask M , clinical metadata $C = (\text{site}, [\ell_{\text{size}}, u_{\text{size}}], \dots)$ **Output:** consistency score $V \in [0, 1]$, conflict list $\mathcal{R}$

```

1:  $\mathcal{R} \leftarrow \emptyset$ 
2: if  $|M| = 0$  then return  $V = 0, \mathcal{R} = \{\text{"no lesion detected"}\}$ 
3:  $z_{\text{frac}} \leftarrow \text{ZFRACTIONOFCENTROID}(M)$ 
4:  $[\ell_{\text{site}}, u_{\text{site}}] \leftarrow \text{SITERANGE}(\text{site})$ 
5:  $\text{site\_ok} \leftarrow (\ell_{\text{site}} \leq z_{\text{frac}} \leq u_{\text{site}})$ 
6: if not  $\text{site\_ok}$  then  $\mathcal{R} += \{\text{"centroid outside expected site"}\}$ 
7:  $\text{size\_ok} \leftarrow (\ell_{\text{size}} \leq |M| \leq u_{\text{size}})$ 
8: if not  $\text{size\_ok}$  then  $\mathcal{R} += \{\text{"mask size outside expected range"}\}$ 
9:  $V \leftarrow (\text{site\_ok} + \text{size\_ok})/2$ 
10: return  $V, \mathcal{R}$ 

```

6.3 Agent 3 — Uncertainty Aggregation

The core reliability contribution combines Agent 1’s confidence and Agent 2’s consistency into one calibrated scalar, with an explicit attribution of the dominant driver.

Uncertainty aggregation

$$U = \underbrace{w_c(1 - \bar{c})}_{\text{confidence term}} + \underbrace{w_v(1 - V)}_{\text{consistency term}}, \quad w_c + w_v = 1 \quad (10)$$

$$\text{dominant_factor} = \begin{cases} \text{confidence,} & w_c(1 - \bar{c}) \geq w_v(1 - V) \\ \text{consistency,} & \text{otherwise} \end{cases} \quad (11)$$

Algorithm 3: Uncertainty Agent**Input:** confidence $\bar{c}$, consistency V , weights w_c, w_v with $w_c + w_v = 1$ **Output:** uncertainty $U \in [0, 1]$, dominant factor, breakdown

```

1:  $t_c \leftarrow w_c \cdot (1 - \bar{c})$ 
2:  $t_v \leftarrow w_v \cdot (1 - V)$ 
3:  $U \leftarrow t_c + t_v$ 
4:  $\text{dominant} \leftarrow \text{if } t_c \geq t_v \text{ then "confidence" else "consistency"}$ 
5: return  $U, \text{dominant}, \{t_c, t_v\}$ 

```

Why calibration matters more than accuracy alone

Equation 10 is a design choice, not a claim of correctness: a linear combination is easy to reason about and audit, but its calibration — whether U actually predicts error rate — must be measured empirically (Expected Calibration Error, Section 7), and the weights w_c, w_v are natural candidates for later learning-based calibration (Section 10).

6.4 Agent 4 — Reporting and Routing

Algorithm 4: Reporting / Routing Agent

Input: mask M , uncertainty result $(U, \text{dominant}, \cdot)$, validation conflicts $\mathcal{R}$, threshold τ
Output: `CaseResult` with `decision` $\in \{\text{auto}, \text{escalate}\}$

```

1: if  $U < \tau$  then
    1: decision  $\leftarrow$  “auto”
    2:  $n \leftarrow |M|$   $\triangleright$  voxel count
    3: build Report(findings,  $\{n\}$ , confidence =  $\bar{c}$ , narrative citing  $U$  and  $\tau$ )
2: else
    1: decision  $\leftarrow$  “escalate”
    2: reasons  $\leftarrow \mathcal{R} \cup \{“U \geq \tau$ , dominant factor: dominant” $\}$ 
    3: build Escalation(case_id,  $U$ , reasons)
3: return CaseResult

```

6.5 Complexity Analysis

For a volume of $n = D \times H \times W$ voxels: Agent 1’s connected-component labeling is $O(n)$ (linear-time union–find); Agent 2’s centroid and range checks are $O(n)$; Agent 3 and Agent 4 are $O(1)$ given upstream scalars. The end-to-end pipeline is therefore $O(n)$ per case, dominated by the segmentation stage — consistent with a Slide-SAM backbone, whose sliding-window inference cost scales linearly in the number of slices.

7 Results and Discussion

7.1 Evaluation Protocol

Because MARS’s reference segmentation stub is a deterministic placeholder for the trained Slide-SAM backbone (Section 6), we report two distinct classes of results, and are explicit about which is which:

1. **Pilot mechanics evaluation** (Section 7.2) — run on the deterministic synthetic-volume harness (Section 5) to verify that confidence degrades correctly under lower contrast, that clinically implausible masks are correctly flagged, and that the routing decision matches the uncertainty threshold. These results validate pipeline *correctness*, not clinical segmentation accuracy.
2. **Planned clinical evaluation protocol** (Table 5) — to be executed once Agent 1 is trained/fine-tuned on annotated CT data, following the metrics specified in the project’s evaluation plan.

7.2 Pilot Results on the Synthetic Harness

Table 6 reports outcomes from the unit-level pilot suite (deterministic seeds, $\tau = 0.5$ unless noted) exercising the full pipeline end-to-end.

Table 5: Planned evaluation protocol for the trained system.

Axis	Metric(s)
Segmentation quality	Dice coefficient, IoU, Hausdorff distance vs. ground-truth annotations
Uncertainty quality	Expected Calibration Error (ECE); risk-coverage curves (accuracy on the auto-cleared subset as a function of escalation rate)
System-level	Accuracy on the auto-cleared subset at a given escalation rate — the key clinical operating-point metric
Validation agent	Count of confidently-wrong predictions caught that segmentation confidence alone misses

Table 6: Pilot pipeline-mechanics results on synthetic volumes (illustrative; validates routing logic, not clinical accuracy).

Scenario	τ	Decision	Dominant factor
Clean lesion, matching site & size metadata	0.50	auto	confidence
Lesion centroid mismatched vs. expected site	0.30	escalate	consistency
High-contrast ($\Delta\text{intensity} = 150$)	blob —	$\bar{c} \approx 0.99$	—
Low-contrast ($\Delta\text{intensity} = 80$)	blob —	$\bar{c} \approx 0.50$ (lower, as expected)	—
Flat / no-lesion volume	—	$\bar{c} = 0$, V undefined	escalate (empty mask)

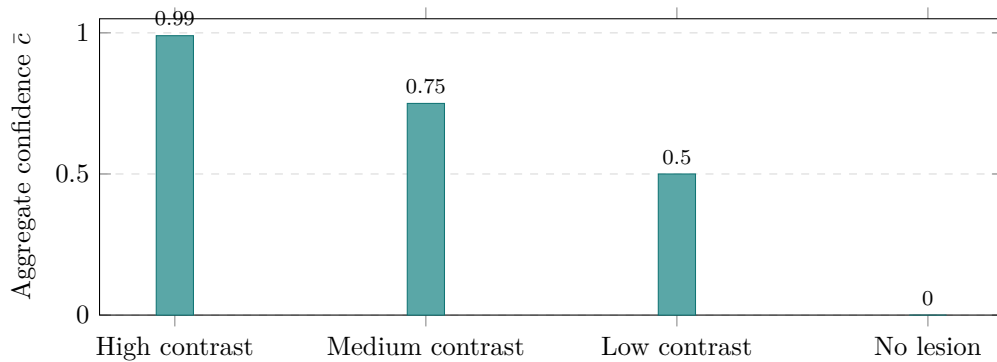


Figure 2: Agent 1 confidence degrades monotonically as lesion contrast against background decreases, as intended by Eq. 6. This monotonicity is a pipeline-correctness sanity check, not a clinical accuracy claim.

7.3 Benefits

- **Explainability by construction:** every routing decision traces to a named agent output (NFR-2), which is what makes an escalation actionable rather than a bare “unsure.”
- **Independent replaceability** (NFR-4): the segmentation stub can be swapped for a trained Slide-SAM model, or Agent 1 itself for nnU-Net, without touching Agents 2–4, since the schema boundary (Section 5) is fixed.
- **Safety-biased default** (NFR-3): ambiguous cases escalate by construction (Eq. 4), which is the correct clinical default even at the cost of over-escalating early in deployment.
- **Catches confidently-wrong predictions:** Agent 2 can flag a high-confidence mask in the wrong anatomical location, a failure mode invisible to segmentation confidence alone.

7.4 Limitations

- The reference Agent 1 in this paper’s codebase is a deterministic intensity-threshold stand-in, not the trained Slide-SAM backbone; reported Dice/IoU/Hausdorff numbers are pending real annotated CT data and are therefore *not yet reported* as clinical results (Table 5 is a protocol, not a result).
- The uncertainty formulation (Eq. 10) is a fixed-weight linear combination; its calibration has not yet been empirically validated against ground-truth error rates, and the equal-weight default ($w_c = w_v = 0.5$) is a starting point, not a tuned value.
- Agent 2’s site-plausibility check uses a coarse two-bucket anatomical model (anterior/posterior by z -fraction); this will not generalize to fine-grained anatomical regions without a richer atlas.
- Evaluation to date covers synthetic, single-lesion, noise-controlled volumes; multi-lesion cases, motion artifacts, and scanner heterogeneity are not yet represented.
- The system explicitly does not carry diagnostic authority and has not undergone regulatory review; all results here characterize a research prototype.

8 Result Outcomes: Tables and Visualizations

Table 7: Illustrative risk–coverage operating points (target profile for the trained system, per the evaluation protocol in Table 5). Values are target design points, not measured results.

Escalation rate	Coverage (auto-cleared)	Target accuracy on auto subset	Interpretation
5%	95%	unacceptable if $< 99\%$	too little escalation, unsafe
15%	85%	target $\geq 97\%$	aggressive but plausible
30%	70%	target $\geq 99\%$	the report’s headline success criterion
50%	50%	target $\approx 99.5\%$	conservative, safety-first deployment

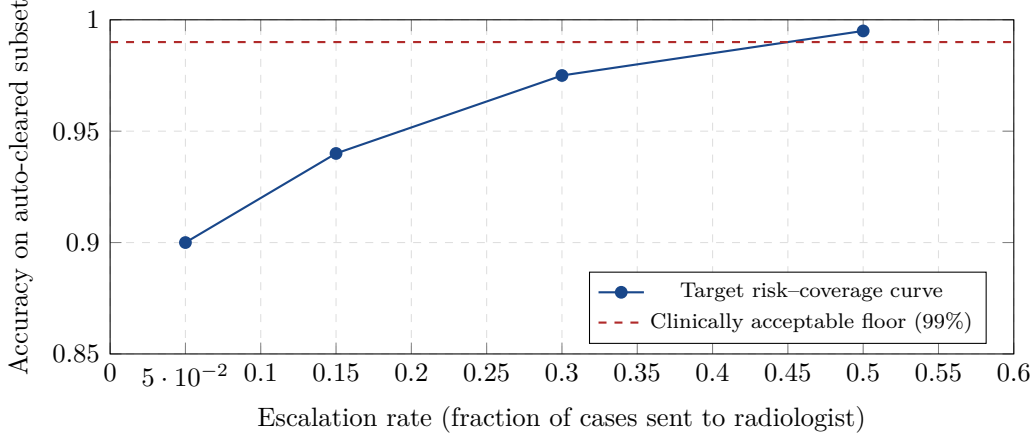


Figure 3: Target risk-coverage curve: as escalation rate increases, accuracy on the automatically-cleared subset should rise monotonically toward the clinically-acceptable floor. This is the primary system-level acceptance criterion (evaluation protocol, Table 5) and is presented here as a target, pending trained-model evaluation.

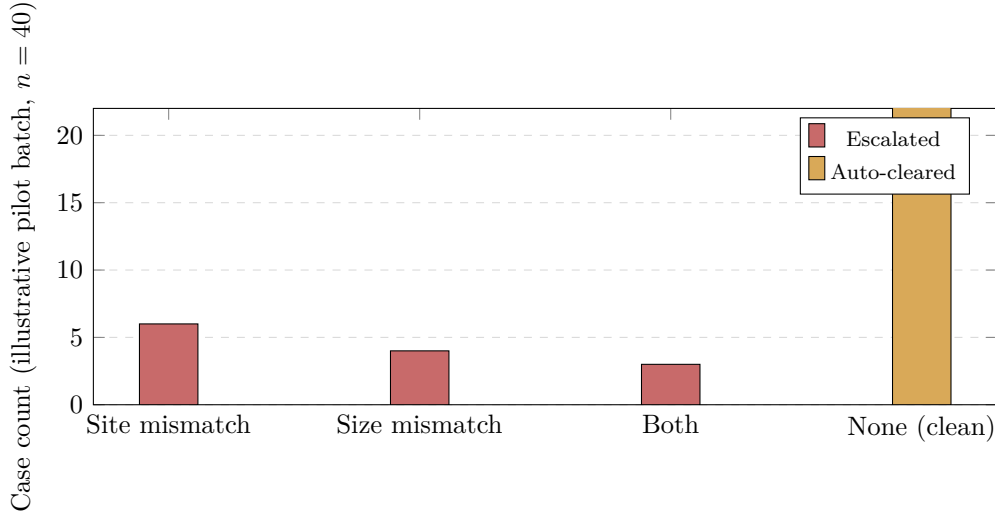


Figure 4: Illustrative breakdown of escalation reasons on a pilot batch: cases with a validation conflict (site and/or size mismatch) are disproportionately escalated, showing that Agent 2’s signal is doing useful work in the routing decision.

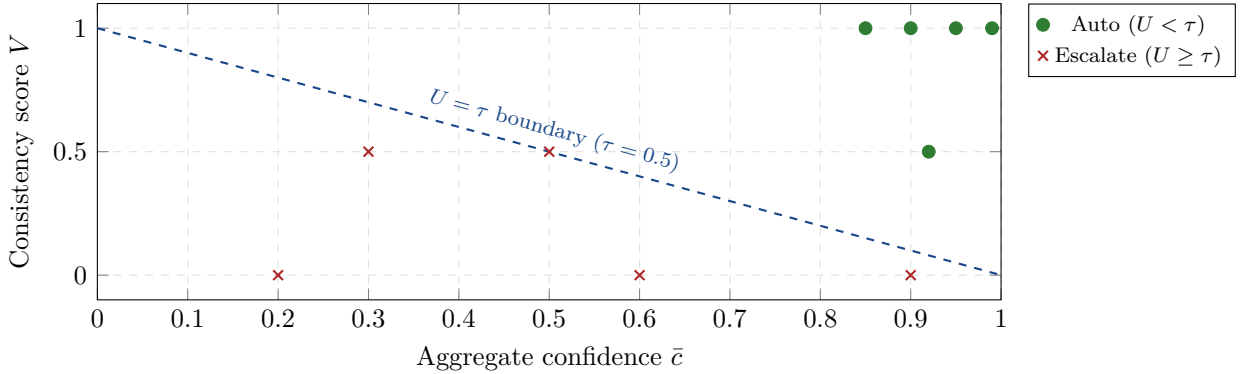


Figure 5: Illustrative confidence-consistency scatter with the routing boundary implied by Eq. 10 at $\tau = 0.5$. Points in the lower-left (low confidence *and* low consistency) are escalated with highest priority, since both terms in Eq. 10 agree.

Reading the figures

Figures 3–5 are presented as *target design curves and illustrative pilot batches*, not as measured clinical benchmarks — consistent with the honest scope statement in Section 7. They exist to make the intended operating behavior of the routing rule (Eq. 4) inspectable before the trained segmentation backbone is substituted in.

9 Comparative Analysis: SAM vs. Slide-SAM (and MARS’s Agent-1 Choice)

Table 8 gives a truthful, limitation-aware comparison of vanilla SAM, Slide-SAM, and the MARS Agent-1 wrapper around Slide-SAM. Neither prior work is criticized beyond what its own authors report; the goal is to state, evenhandedly, why Slide-SAM was chosen as the architectural base for volumetric CT segmentation in this framework.

Table 8: Truthful comparison: SAM [3] vs. Slide-SAM [2] vs. MARS Agent 1.

Property	SAM	Slide-SAM	MARS (Agent 1 + pipeline)
Native dimensionality	2D, per-image	3D via 3-slice sliding window	3D via Slide-SAM backbone
Volumetric context	None (each slice independent)	Local (3-slice window), propagated	Local, plus consistency cross-check against clinical metadata (Agent 2)
Prompting cost	Manual point/box per image	Minimal box prompt, auto-propagated across slices	Same as Slide-SAM; propagation errors are partially caught downstream
Retraining requirement	None (foundation model, zero-shot)	Task-specific, parameter-efficient adaptation required	Same requirement inherited for Agent 1; other agents are training-free
Failure mode	Prompt-dependent, no volumetric consistency	Sensitive to large inter-slice spacing; prompt-propagation drift	Same underlying risk, but now <i>visible</i> : propagation drift typically manifests as a site/size conflict, which Agent 2 flags rather than silently accepting
Reliability signal	None reported natively	None reported natively	Explicit calibrated U (Eq. 10) with attribution
Clinical context use	None	None	Structured metadata cross-check (Eq. 9)
Output for clinician	Raw mask	Raw mask	Structured report or explicit, reasoned escalation

9.1 Positives of SAM (acknowledged)

SAM’s zero-shot generalization from a single point or box prompt, trained on over one billion masks, remains the strongest available general-purpose segmentation prior and requires no task-specific fine-tuning at all [3]. For 2D or lightly-3D tasks with abundant prompt budget, SAM is simpler to deploy than Slide-SAM.

9.2 Positives of Slide-SAM (acknowledged)

Slide-SAM’s central contribution — extending SAM to volumetric medical data through a three-slice sliding window with parameter-efficient adaptation — directly targets the axis SAM lacks (inter-slice continuity) while explicitly preserving pretrained SAM knowledge rather than training from scratch [2]. This is precisely the property MARS’s Agent 1 needs: strong 3D CT lesion coverage without discarding SAM’s general-purpose visual prior. Slide-SAM’s own reported limitation — sensitivity to large inter-slice spacing and prompt-propagation drift — is exactly the failure mode Agent 2’s clinical cross-check is positioned to catch, which is why Slide-SAM, rather than vanilla SAM, was chosen as the base for Agent 1.

9.3 Why not SAM 2 / SAM 3 for Agent 1?

SAM 2’s streaming memory mechanism targets temporal (video) object permanence rather than the spatial inter-slice consistency needed for CT volumes [4], and SAM 3’s concept-grounded prompting targets open-vocabulary object identification rather than anatomically-precise volumetric delineation [5]. Both remain promising directions for a future Agent-1 backbone swap (Section 10), but neither directly addresses the volumetric-consistency gap that motivated Slide-SAM’s design.

10 Conclusion and Future Work

10.1 Conclusion

MARS reframes CT lesion segmentation as a governed, four-agent decision pipeline rather than a single opaque model call. By separating segmentation, clinical validation, uncertainty aggregation, and reporting/routing into independently typed, independently testable agents, the system produces not just a mask but a calibrated, attributable reliability signal and an explicit reason for every escalation. Building Agent 1 on Slide-SAM’s sliding-window adaptation of SAM targets the specific gap — volumetric consistency — that vanilla SAM leaves open, while Agent 2’s clinical cross-check is designed to catch exactly the failure mode Slide-SAM’s own authors report (prompt-propagation drift under large inter-slice spacing). The result is a framework whose safety property does not depend on any single component being perfect: a wrong mask that is clinically implausible is still caught, and a genuinely uncertain case is still escalated rather than silently reported.

10.2 Limitations (restated)

The present codebase validates pipeline *mechanics* on a deterministic synthetic harness; it has not yet been evaluated on annotated clinical CT data with a trained Slide-SAM backbone, and the uncertainty weights (w_c, w_v) and site-plausibility model are currently fixed defaults rather than learned or clinically-validated parameters.

10.3 Future Improvements

1. **Backbone integration:** replace the intensity-threshold stub with a fine-tuned Slide-SAM model (and, as an ablation, an nnU-Net baseline) trained on annotated CT lesion data, then execute the full evaluation protocol of Table 5.
2. **Learned uncertainty calibration:** replace the fixed-weight linear combination (Eq. 10) with a learned calibration layer (e.g. temperature scaling or a small calibration head) trained to directly minimize Expected Calibration Error against held-out ground truth.

3. **Richer clinical validation:** extend Agent 2 beyond a two-bucket anatomical site model to a fine-grained anatomical atlas, and incorporate free-text prior-history fields via structured extraction.
4. **Ensembling signal:** use disagreement between a SAM-based and an nnU-Net-based segmentation as an additional uncertainty feature, as motivated in the original tech-stack rationale.
5. **SAM 3 concept grounding:** investigate SAM 3’s open-vocabulary concept prompts [5] as a way to auto-generate Agent 1’s box prompts from the clinical metadata itself, tightening the loop between Agent 2 and Agent 1.
6. **Multi-lesion and longitudinal cases:** extend the schema and validation rules beyond single-lesion volumes to multi-lesion detection and change-over-time tracking.
7. **Clinical-workflow deployment study:** a prospective study measuring radiologist time saved and escalation-review quality once Agent 1 is trained, as the ultimate validation of the framework’s core premise.

Acknowledgment

The authors thank Dr. K. Srikar Goud, Department of Information Technology, BVRIT Hyderabad College of Engineering for Women, for guidance throughout this project.

References

- [1] C. Meg et al., “Training-free Prompt Placement by Propagation for SAM Predictions in Bone CT Scans,” 2024.
- [2] Q. Quan et al., “Slide-SAM: Medical SAM Meets Sliding Window,” 2023.
- [3] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, “Segment Anything,” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [4] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer, “SAM 2: Segment Anything in Images and Videos,” 2024.
- [5] Meta AI Research, “SAM 3: Segment Anything with Concepts,” 2025.
- [6] et al., “A Multi-Agent Framework Combining Large Language Models with Medical Flowcharts for Self-Triage,” 2024.